

CPTG Fraction-of-50 Diagnostic Catalogue

Revised by subject; diagnostic comments only; strongest entries first within each subject

Prepared for Curvature Polarization Transport Gravity (CPTG). Updated May 7, 2026.

Scope and interpretation rules

This catalogue records denominator-50 and nearby compact-rational structures. It separates CPTG equation-level constants from dimensionless diagnostics, dynamical solar-system structure, rounded catalog quantities, and operational run settings. External entries are comparative gravitational diagnostics only; they are not evidence for CPTG unless they survive model-specific ablation, null testing, and independent reruns.

Denominator-50 lattice: $x_n = n/50$, n in $\mathbb{Z}$.

Tier	Meaning	Use in this document
Strong	Equation-level CPTG constants or real external dynamical structures.	Can be retained as primary diagnostic material.
Medium	Dimensionless physical ratios, cross-validated geometry, or stable host-normalized relations.	Retain, but do not treat as theory evidence by itself.
Weak	Rounded AU distances, rounded statistics, conventional units, or reporting grids.	Watch-list only.
Context	Background numerical recurrence such as 50-60 e-folds or circular constants.	Do not use as scoring target.

Items are ordered from strongest to weakest inside each subject. Diagnostic comments are intentionally short. Unit-dependent precession entries are not used.

1. CPTG equation-level fraction structure

Primary CPTG entries are ranked above all external analogies because the fractions occur in the model equations or fixed definitions.

Item	Value	Fraction-of-50 form	Required diagnostic comment	Strength	Source
Suppression exponent	$123/50 = 2.46$	123/50	Equation-level response exponent.	Strong	[CPTG]
Suppression power	$11/50 = 0.22$	11/50	Equation-level response power.	Strong	[CPTG]
Gas source weight	27/50	27/50	Equation-level baryonic source coefficient.	Strong	[CPTG]
Disk source weight	23/50	23/50	Equation-level baryonic source coefficient.	Strong	[CPTG]
Gas-disk closure	$27/50 + 23/50 = 1$	50/50	Closed complementary source pair.	Strong	[CPTG]
Gas-disk midpoint	$(27/50 + 23/50)/2$	25/50	Central local-source scale.	Strong	[CPTG]
Bulge source weight	$32/50 = 16/25$	32/50	Compact-source coefficient; separate from gas-disk pair.	Strong	[CPTG]
Geometric remnant exponent	$7/5 = 1.4$	70/50	Equation-level radial-shape power.	Strong	[CPTG]
Geometric denominator exponent	$12/5 = 2.4$	120/50	Equation-level radial-shape power.	Strong	[CPTG]
Transport coefficient	$\eta = 1/2$	25/50	Equation-level averaging/transport coefficient.	Strong	[CPTG]
Structural-mode lower cutoff	$R/5$	10/50 of R	Explicit interval in structural-mode definition.	Medium	[CPTG]
Outer attractor	$s_{\infty} = 1/4$	quarter-power	Compact fraction, not integer n/50.	Medium	[CPTG]
Scale coefficient	$3/(16\pi) = 0.05968$	near 3/50	Circular-normalized scale relation; not exact n/50.	Weak-to-medium	[CPTG]

2. CPTG CMB and early-transport diagnostics

These entries are model-side diagnostics from the current source-proxy program. Geometry and locked-ratio definitions rank above rounded fit statistics.

Item	Value	Fraction-of-50 form	Required diagnostic comment	Strength	Source
Locked shell ratio	$8/25 = 0.32$	16/50	Active constrained shell diagnostic.	Strong	[CMB script]
Ratio scan grid	0.30, 0.32, 0.34, 0.36	15/50-18/50	Controlled diagnostic scan; not optimized as evidence.	Medium	[CMB script]
Recombination/equality ratio	$(1+z^*)/(1+z_{eq}) \sim 0.3203$	near 16/50	Early-expansion timing ratio.	Medium	[CMB]
Raw CMB shell geometry	$178.370773 / 552.181947 = 0.3230$	0.32-class	Dimensionless shell width/center ratio.	Medium	[CMB]
CV CMB shell geometry	$176.977272 / 539.719208 = 0.3279$	0.32-class to 1/3	Cross-validated shell geometry.	Medium	[CMB]
Matter/radiation at recombination	$(1+z_{eq})/(1+z^*) = 3.1222$	near 156/50	Close to pi; timing composition diagnostic.	Medium	[CMB]
Phi watch: inverse phi	$1/\phi = 0.618034$; grid = 31/50	31/50	Watch only; do not optimize.	Weak-to-medium	[CMB script]
Phi watch: inverse phi^2	$1/\phi^2 = 0.381966$; grid = 19/50	19/50	Watch only; do not optimize.	Weak-to-medium	[CMB script]
Phi watch: phi	$\phi = 1.618034$; grid = 81/50	81/50	Watch only; do not optimize.	Weak-to-medium	[CMB script]
TT/TE/EE holdout scores	TT 7.7774; TE 7.3556; EE 2.0048	near grid values	Weak because statistics can grid-align by rounding.	Weak	[CMB]
50M CMB run checkpoint	50,000,000 candidates; best Delta chi^2 = 21.1850	50M operational	Run-state note; not a physics diagnostic.	Operational	[Checkpoint]

3. External cosmology and circular-normalization context

These are not CPTG constants. They are retained only to track whether the same compact numerical classes recur in independent cosmological quantities.

Item	Value	Fraction-of-50 form	Required diagnostic comment	Strength	Source
CDM physical density	$\Omega_c h^2 \sim 0.120$	6/50	External Planck-like parameter.	Medium	[Planck]
Matter physical density	$\Omega_b h^2 + \Omega_c h^2 \sim 0.1425$	near 7/50	External derived parameter.	Medium	[Planck]
Matter density fraction	$\Omega_m \sim 0.315$	near 16/50	External cosmological ratio.	Medium	[Planck]
Dark-energy fraction	$\Omega_\Lambda \sim 0.685$	near 34/50	Complementary external ratio.	Medium	[Planck]
Baryon/matter fraction	~ 0.157	near 8/50	External composition ratio.	Medium	[Planck]
CDM/matter fraction	~ 0.843	near 42/50	External composition ratio.	Medium	[Planck]
MOND acceleration coefficient	a_0 coefficient ~ 1.2	60/50	External modified-gravity scale coefficient.	Weak-to-medium	[MOND]
π	3.14159	near 157/50	Circular-normalization benchmark.	Context	[Math]
2π	6.28318	near 314/50	Circular-normalization benchmark.	Context	[Math]
$1/(2\pi)$	0.15915	near 8/50	Inverse circular benchmark.	Context	[Math]
Inflation e-fold context	$N_{\text{obs}} \sim 50\text{-}60$	direct 50-scale	Context only; model-dependent.	Context	[Inflation]

4. Solar-system dynamical structure

Dynamical structures are the strongest external solar-system entries. They are useful guardrails because ordinary gravitational mechanics naturally produces compact rational structure.

4.1 Lagrange points

Item	Value	Fraction-of-50 form	Required diagnostic comment	Strength	Source
L4/L5 stability threshold	small/large $< 1/24.96 = 0.04006$	near 2/50	Dimensionless stability boundary.	Strong	[NASA Lagrange]
Stable points / total points	2 stable / 5 total	20/50	Count ratio; secondary.	Medium	[NASA Lagrange]
Triangular point angle	60 deg = $\pi/3 = 1.0472$ rad	near 52/50 rad	Circular geometry; unit-sensitive in degrees.	Weak-to-medium	[NASA Lagrange]

4.2 Asteroid belt and co-orbital populations

Item	Value	Fraction-of-50 form	Required diagnostic comment	Strength	Source
Kirkwood gap: 3:1 Jupiter resonance	3.0	150/50	Mean-motion resonance gap.	Strong	[JPL Kirkwood]
Kirkwood gap: 5:2 Jupiter resonance	2.5	125/50	Mean-motion resonance gap.	Strong	[JPL Kirkwood]
Kirkwood gap: 2:1 Jupiter resonance	2.0	100/50	Mean-motion resonance gap.	Strong	[JPL Kirkwood]
Hilda asteroids	3:2 Jupiter resonance	75/50	Stable resonant population.	Strong	[NASA NTRS/ADS]
Jupiter Trojans	1:1 co-orbital resonance	50/50	L4/L5 co-orbital population.	Strong	[NASA NTRS]
Kirkwood gap: 7:3 Jupiter resonance	2.333...	near 117/50	Listed resonance; not exact $n/50$.	Strong-to-medium	[JPL Kirkwood]
Ceres + Vesta mass share	about 40% of asteroid-belt mass	20/50	Mass-concentration statistic.	Medium	[NASA Dawn]
Belt center near 2.5 AU	about 2.5 AU	125/50	Rounded location; weak.	Weak	[JPL]

4.3 Saturn ring structure

Ring resonances and tidal boundaries rank above ring-edge radii because they encode dynamics rather than naming boundaries.

Item	Value	Fraction-of-50 form	Required diagnostic comment	Strength	Source
Classical Roche limit	2.44 Saturn radii	122/50	Tidal disruption/accretion boundary.	Strong	[Roche]
Cassini Division / Mimas	2:1 resonance	100/50	Dynamical ring-clearing resonance.	Strong	[NASA/USGS]
Cassini Division width	0.0780 Saturn radii	near 4/50	Ring-width diagnostic.	Medium	[USGS Rings]
A ring width	0.2423 Saturn radii	near 12/50	Ring-width diagnostic.	Medium	[USGS Rings]
B ring width	0.4231 Saturn radii	near 21/50	Ring-width diagnostic.	Medium	[USGS Rings]
C ring width	0.2904 Saturn radii	near 15/50	Ring-width diagnostic.	Medium-low	[USGS Rings]
Roche Division width	0.0431 Saturn radii	near 2/50	Narrow division; weaker than resonance.	Medium-low	[USGS Rings]
G ring width	0.1327 Saturn radii	near 7/50	Broad ring-width watch.	Weak-to-medium	[USGS Rings]

4.4 Planetary mean-motion and near-resonance entries

Item	Value	Fraction-of-50 form	Required diagnostic comment	Strength	Source
Pluto-Neptune resonance	3:2 = 1.5	75/50	Real mean-motion resonance.	Strong	[NASA Pluto/KBO]
Jupiter-Saturn near resonance	$P_S / P_J = 29.447498 / 11.862615 = 2.4823$	near 125/50	Near 5:2 period structure; not exact lock.	Strong-to-medium	[JPL Planets]
Uranus/Neptune period ratio	$84.016846 / 164.79132 = 0.5098$	near 25/50	Outer-planet period ratio.	Medium	[JPL Planets]
Venus/Earth period ratio	$0.615197 / 1.000017 = 0.6152$	near 31/50	Phi-watch overlap.	Medium	[JPL Planets]
Mars/Earth period ratio	$1.880848 / 1.000017 = 1.8808$	near 94/50	Planet-period ratio.	Medium	[JPL Planets]
Mercury/Earth period ratio	$0.240847 / 1.000017 = 0.2408$	near 12/50	Planet-period ratio.	Medium	[JPL Planets]
Rounded AU distances	e.g., 0.39, 0.72, 1.52 AU	various n/50	Weak because AU entries are rounded and convention-dependent.	Weak	[JPL/NASA]

5. Planet-moon and moon-moon structure

Host-normalized size/GM ratios and moon-moon period ratios are stronger than raw moon counts. Discovery-limited moon counts are not used.

5.1 Host-normalized planet-moon ratios

Item	Value	Fraction-of-50 form	Required diagnostic comment	Strength	Source
Charon / Pluto radius	0.50997	near 25/50	Clean outer-system double-body ratio.	Medium-high	[JPL Planets/Sats]
Charon / Pluto GM	0.12205	near 6/50	Dimensionless gravity-parameter ratio.	Medium-high	[JPL Sats]
Moon / Earth volume	0.02028	near 1/50	Physical volume ratio.	Medium	[NASA Moon]
Moon / Earth radius	0.27270	near 14/50	Host-normalized size ratio.	Medium	[JPL Planets/Sats]
Moon / Earth surface gravity	0.16560	near 8/50	Physical surface-gravity ratio.	Medium	[NASA Moon]
Moon / Earth escape velocity	0.21229	near 11/50	Physical escape-velocity ratio.	Medium	[NASA Moon]
Triton / Neptune radius	0.05493	near 3/50	Host-normalized size ratio.	Medium	[JPL Planets/Sats]
Titan / Saturn radius	0.04422	near 2/50	Host-normalized size ratio.	Medium	[JPL Planets/Sats]
Ganymede / Jupiter radius	0.03764	near 2/50	Host-normalized size ratio.	Medium	[JPL Planets/Sats]
Callisto / Jupiter radius	0.03448	near 2/50	Host-normalized size ratio.	Medium	[JPL Planets/Sats]
Europa / Jupiter radius	0.02233	near 1/50	Host-normalized size ratio.	Medium	[JPL Planets/Sats]
Ariel / Uranus radius	0.02283	near 1/50	Host-normalized size ratio.	Medium	[JPL Planets/Sats]
Umbriel / Uranus radius	0.02305	near 1/50	Host-normalized size ratio.	Medium	[JPL Planets/Sats]
Moon / Earth GM	0.01230	below 1/50	Completeness entry; weaker than volume.	Weak-to-medium	[JPL Sats]

5.2 Moon-moon orbital-period structure

System	Relationship	Value	Fraction-of-50 form	Required diagnostic comment	Strength	Source
Jupiter	Europa / Io	1.999999	100/50	Io-Europa resonance-family entry.	Strong	[JPL Sat Elem]
Jupiter	Io-Europa-Ganymede	1:2:4 family	50/50, 100/50, 200/50	Laplace resonance-family structure.	Strong	[JPL Sat Elem]
Saturn	Tethys / Mimas	2.00314	near 100/50	Near 2:1 moon-moon structure.	Strong	[JPL Sat Elem]
Saturn	Dione / Enceladus	1.99743	near 100/50	Near 2:1 moon-moon structure.	Strong	[JPL Sat Elem]
Saturn	Hyperion / Titan	1.33434	near 67/50	Near 4:3 resonance-class entry.	Strong-to-medium	[JPL Sat Elem]
Jupiter	Callisto / Ganymede	2.33250	near 117/50	Compact ratio; weaker than locked pair.	Medium	[JPL Sat Elem]
Pluto	Hydra / Charon	5.98069	near 299/50	Near 6:1 Charon-normalized watch.	Medium	[JPL Sat Elem]
Pluto	Kerberos / Charon	5.03662	near 252/50	Near 5:1 Charon-normalized watch.	Medium	[JPL Sat Elem]
Pluto	Nix / Charon	3.89058	near 195/50	Near 4:1 Charon-normalized watch.	Medium	[JPL Sat Elem]
Pluto	Styx / Charon	3.15630	near 158/50	Pi-class Charon-normalized watch.	Medium	[JPL Sat Elem]
Uranus	Titania / Umbriel	2.10075	near 105/50	Compact period ratio; not primary.	Weak-to-medium	[JPL Sat Elem]

6. Weak watch-list entries

Weak entries remain useful only as pattern-watch context. They should not drive parameter choices or public claims.

Subject	Item	Value	Fraction-of-50 form	Required diagnostic comment	Strength	Source
Planet distances	Mercury distance	~0.39 AU	near 20/50	Rounded AU value.	Weak	[NASA scale guide]
Planet distances	Venus distance	~0.72 AU	36/50	Rounded AU value.	Weak	[NASA scale guide]
Planet distances	Mars distance	~1.52 AU	76/50	Rounded AU value.	Weak	[NASA scale guide]
Adjacent planets	Mercury/Venus distance ratio	$0.39/0.72 = 0.5417$	near 27/50	Rounded-input ratio.	Weak-to-medium	[NASA scale guide]
Adjacent planets	Jupiter/Saturn distance ratio	$5.20/9.54 = 0.5451$	near 27/50	Rounded-input ratio.	Weak-to-medium	[NASA scale guide]
Moon orbit	Moon orbital eccentricity	0.0554	near 3/50	Single orbital-shape parameter.	Weak-to-medium	[JPL Sat Elem]
Pluto orbit	Perihelion/aphelion distance ratio	$30/49.3 = 0.6085$	near 30/50	Rounded Pluto distance ratio.	Weak-to-medium	[NASA Pluto]
Pluto orbit	Average/aphelion distance ratio	$39/49.3 = 0.7911$	near 40/50	Rounded Pluto distance ratio.	Weak-to-medium	[NASA Pluto]

7. Required ablation and validation comments

Area	Required diagnostic handling
CPTG constants	Perturb each equation-level denominator-50 coefficient by one lattice step and report benchmark change.
Gas-disk pair	Test closure preserved and closure broken.
Bulge coefficient	Test independently from gas-disk pair.
SPARC benchmark	Recompute margins against MOND and MOND-plus-Upsilon for each ablation.
CMB source-proxy models	Keep denominator-50 tracking diagnostic-only until final model revision.
CMB shell geometry	Report nearest-lattice distance for dimensionless ratios, not only score values.
CMB validation	Use TT/TE/EE consistency, leave-one-spectrum-out validation, block holdouts, and full-pipeline nulls.
Solar-system diagnostics	Treat resonances as real external gravitational structure, not CPTG evidence.
Weak rounded entries	Retain only as watch-list examples.
Phi/Fibonacci watch	Track 31/50, 19/50, 81/50, and adjacent-ratio resonances as diagnostic-only.

Research interpretation

The strongest denominator-50 structures remain the constants embedded directly in CPTG equations. The strongest external comparisons are real dynamical structures such as resonances, stability thresholds, and tidal boundaries. Dimensionless host-normalized ratios are useful medium-strength diagnostics. Rounded AU distances, rounded statistical outputs, and operational run counts are weak or non-physical watch-list items.

Development rule: observe and ablate first; constrain only in the final model revision.

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